

AI-Powered Weed Seed Recognition and Filtering Platform for Smart Rice Farming

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Abstract—The presence of weed seeds, especially weedy rice, in post-harvest rice grain samples represents a significant risk to seed purity, germination quality, and marketability. Traditional methods of manual and mechanical seed separation are time-consuming, inconsistent, and often inadequate to discriminate against visually similar impurities. This paper presents the design and development of an AI-powered filtering and classification system for seeds based on a hybrid deep learning model that merges ConvNeXt with a new Convolutional Neural Network (CNN). The system was trained on a custom-labeled dataset of rice and weed seed images and achieved approximately 93% accuracy in classification. It enables real-time detection, filtering, and contamination reporting through a mobile-friendly dashboard, addressing the most important challenges in rice quality assurance. The integration of image preprocessing, model prediction, and visual feedback transforms manual inspection into an automated, scalable, and accurate process. It helps transform agricultural post-harvest operations by offering a reliable tool for rice millers, quality control officers, and farmers to ensure grain purity and improve market readiness.

Keywords—Agriculture Technology, AI, Deep Learning, Rice Farming, Mobile Application, Weed Seed Diagnosis

I. INTRODUCTION

The international market demand for high-quality, contaminant-free rice brings to the forefront the need for intelligent post-harvest quality management systems. One of the most significant problems in rice production is the contamination of grain samples with weed seeds—particularly weedy rice (*Oryza sativa* f. *spontanea*) and other grassy weeds. These contaminants not only destroy the commercial quality of rice but also jeopardize seed purity and agricultural sustainability. Traditional seed identification methods for weeds, relying principally on manual inspection and mechanical grading, are prone to inefficiencies, inconsistencies, and scalability limitations.

Advances in artificial intelligence (AI) and computer vision—especially deep learning architectures like Convolutional Neural Networks (CNNs)—have created opportunities to revolutionize agricultural processing. This research explores the development of a hybrid deep learning model by in- targeting ConvNeXt with a custom-designed CNN for the automatic identification and filtering of weed seeds from rice grain samples. The system is integrated with real-time image classification, a mechanical filtering module, and a mobile- accessible app, allowing practical deployment in field and industrial environments.

Unlike earlier efforts that have focused primarily on in-field weed detection or general seed grading, the proposed system addresses the urgent need for post-harvest weed seed identification and removal. The system combines mobile access, real-time feedback loops, and domain-expert collaboration and hence presents itself as a multi-purpose tool for farmers, rice millers, and quality inspectors. Through this work, we are contributing not only to the automation of seed purity analysis but to the general vision of precision agriculture and sustainable food production.

II. LITERATURE REVIEW

Weed infestation in rice grain is a classic agronomic problem that has a direct bearing on seed purity, crop yield, and market value. Traditional post-harvest detection and sorting methods of weed seeds utilize visual examination and physical separation, which are time-consuming, manual, and prone to human errors. The emergence of artificial intelligence (AI) in the guise of computer vision has introduced revolutionary changes in implementing agriculture quality control automa- tion. This literature review detects significant contributions and gaps in applying machine learning and deep learning methods to seed classification and detection of weeds.

Ruslan et al. [1] demonstrated an initial use of a machine vision classifier to distinguish between weedy and cultivated rice using color and shape features aided by IMAQ Particle Filter 3 to reduce noise. Their work confirmed the importance of image pre-processing to increase the rate of classification. Dadashzadeh and Abbaspour-Gilandeh [2] also designed a stereo vision-based weed monitoring system applying Gaussian and edge-detection filters but for farm conditions and not for post-harvest seed filtering.

Among the most recent successes in seed classification was that of Kiratiratanapruk and Temniranrat [3], who proposed a grading system based on geometric and statistical features using a 2,900-image database of seeds. While useful, their system did not include real-time filtering or expert-in-the-loop capability. Murad et al. [4] undertook a more extensive survey wherein they investigated deep models such as GoogleNet and ResNet for weed identification and concluded that the majority of the applications were aimed at in-field weed management rather than post-harvest sorting.

Sohail et al. [5] suggested a total automation of seed categorization by morphological characteristics and an SVM classifier with over 95% accuracy. However, the system was not suggested for real-time operation with filtration equipment or mobile devices. Haq et al. [6] used transfer learning coupled with Gabor wavelet filtering to cope with varying illumination and scale in images of rice crops, but their application remained field-oriented and not directly usable in post-harvest quality assessment.

Collectively, the current works shed light on the growing potential of AI systems in agricultural contexts while also demonstrating substantial limitations. Most systems are focused at the field-stage weed detection, lack adaptive learning, and lack mobile interfaces as well as expert feedback mechanisms. The current research closes the gaps by making use of a hybrid deep learning framework based on a ConvNeXt-based feature extractor, a tailored CNN architecture, real-time mechanical filtering, and Firebase-based performance monitoring. Besides, the system is designed to be mobile platform accessible and capable of working with diverse rice varieties and conditions and is hence an adamant solution for post-harvest seed cleaning in resource-poor as well as industrial setups.

III. METHODOLOGY

The smart weed seed filtering system for rice was realized in a multi-phased approach that consisted of data collection, developing a deep learning model, system development, and deployment. The solution presented hereby exploits a hybrid deep learning model based on ConvNeXt and a novel Convolutional Neural Network CNN to accurately classify rice and weeds seeds from input images. Below is an overview of each phase in the research process, starting with requirements analysis, implementation, and evaluation.[12]

A. Architecture

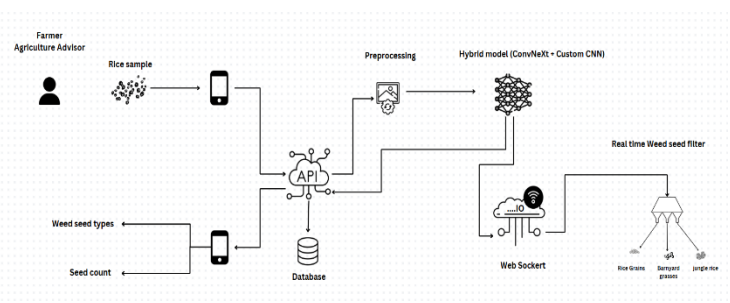


Fig. 1. Overall system diagram

The proposed system is a smart agriculture modulated system that automatically detects and filters weed seeds from post-harvest rice grain samples. The overall system architecture, as illustrated in Fig. 1, consists of five layers integrated together, namely user interface, preprocessing pipeline, classification engine, real-time communication module, and cloud storage.

User Access & Roles: The system supports role-based login for two principal user categories—Farmers, who are able to upload seed images and view purity reports, and Agricultural Advisors, who are able to view farmer inputs and offer professional recommendations.

Image Capture & Preprocessing: Rice seed images are uploaded through users photographed against uniform backgrounds. Users process images through operations such as brightness normalization, noise removal (Gaussian blur), removal of background, and edge detection—to make them transparent and uniform for further analysis. The core of the system is a hybrid deep learning framework that includes ConvNeXt (used as the pretrained feature extractor) and a proprietary CNN classifier. Each seed in an image is labeled and segmented into one of several classes: rice, barnyardgrass, jungle rice, Saromacca grass, glume, or jungle rice B. Results are graphically overlaid on the original image using bounding boxes and color-coded marks.

Real-Time Filtering Integration: With Socket.IO, the system provides instructions to the seed-sorting machinery in real-time for real-time weeding seed removal. Real-time hardware feedback loops and on-the-spot physical filtering of impurities are made possible.

Output & Cloud Storage: Processed results—overall seeds number, weed seed distribution, and purity rate—are stored securely in cloud databases (MySQL or MongoDB). Users can view historical reports, download reports, and track performance history. Feedback and alerts are given in real time via Firebase Cloud Messaging (FCM) for increased interactivity and traceability.

B. AI Model Design and Implementation

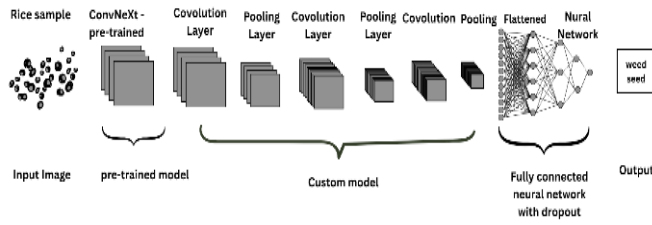


Fig. 2. Model Architecture

The foundation of the proposed weed seed identification system is rooted in deep learning, which is a field in artificial intelligence that enables machines to learn hierarchical representations from data. In this research, deep learning offers a method of accurate classification of rice and weed seeds by mimicking human-level vision through image-based pattern recognition. Fig. 2 depicts the simplified diagram of the Model architecture

Deep Learning Foundations: Deep learning models are based on artificial neural networks (ANNs)—multi-layered computer networks inspired by the human brain. Deep learning models leverage enormous volumes of data and iterative training via optimization algorithms like Adam or Stochastic Gradient Descent (SGD) to reduce errors in their predictions. This technique is very effective in applications such as object detection, image classification, and visual recognition—and is especially well-suited to rice seed quality examination.[11]

Deep learning replaces manual, time-consuming sorting using traditional techniques in this setup and replaces it with an automated and scalable classification pipeline that is able to identify cultivated and contaminated weed seeds in real-time.

Hybrid Deep Learning Architecture: A hybrid deep learning model was used for the sake of better accuracy and computational effectiveness. This is made up of a pre-trained ConvNeXt network integrated with a custom-made Convolutional Neural Network (CNN).

ConvNeXt for mobile and embedded vision applications inserts depth wise separable convolutions, reducing model size and inference time with little impact on accuracy. It is a very good feature extractor due to transfer learning from the ImageNet dataset. The extracted features are passed into a custom CNN branch, which is fine-tuned separately for rice and weed seed morphology.

Model Architecture Overview:

- **Base Layers:** General-purpose pre-trained ConvNeXt (frozen to selected depth) for feature extraction.
- **Custom CNN Layers:** Extra convolutional layers with ReLU activation, pooling, and dropout to learn task-specific features.
- **Fully Connected Layers:** Dense layers to consolidate

extracted features and perform final multi-class classification using SoftMax activation.

The architecture accommodates classification over multiple seed categories, such as:

- Cultivated rice seeds (Samba, Nadu, Kakulu, etc.)
- Weed seeds (Barnyardgrass, Jungle Rice, Saromacca Grass, Glume, Jungle Rice B)

Convolutional Neural Networks (CNN): CNNs form the building blocks of the classification process. They comprise:

- **Convolutional Layers:** Pick out local features like edges, textures, and intricate seed features.
- **Pooling Layers:** Down sample feature maps to reduce computation with minimal spatial information loss.
- **Dropout Layers:** Prevent overfitting during training.
- **Fully Connected Output:** Final SoftMax layer for multi-class probability prediction.

CNN elements are utilized in both the transfer learning phase (through ConvNeXt) and in the custom layers tailored for accuracy in the specific domain. Through the combination of transfer learning and custom fine-tuning, the model attains enhanced convergence rate and high accuracy in discriminating morphologically akin seeds.

C. Real-Time Filtering and WebSocket Integration

One of the most vital components of the proposed system is its ability to translate AI-driven seed classification outcomes into mechanical filtering operations in real time. The moment that an image of a rice sample is captured and processed through the deep learning model, the system gives a total count of all the types of seeds identified, such as cultivated rice grains and weed seed impurities.

For the activation of automatic physical sorting, the system uses a shelled filtering mechanism, where each shell depicts a specific kind of weed seed (i.e., Jungle Rice, Barnyardgrass, Saromacca Grass). Based on the classification phase result, the system identifies the seed type that is most dominant in the infested sample. For instance, when the count of Jungle Rice seeds exceeds a predetermined point, the filtering nozzle positively deflects the flow to the Jungle Rice shell so that these impurities are physically ejected from the rice grain flow.

This intelligent decision-making is brought about by WebSocket communication, which has been established using Socket.IO. This real-time interface establishes a two-way communication channel between the mechanical seed filtering hardware and the AI server. The AI backend communicates real-time sorting commands to the hardware controller after seed classification, which then actuates the respective filtering nozzle to expel the targeted weed seeds.

Major functionalities include:

- **Dynamic Sorting Control:** Filtering pathway is computed in real time by the system based on seed counts.
- **Seamless Hardware Feedback Loop:** Socket.IO provides millisecond-scale latency, enabling near-instant response by

the filtering equipment.

- **Remote Monitoring and Intervention:** Live filtering activities can be monitored by processing technicians or agricultural consultants and manually intervened in case irregularities are detected.
- **Scalability and Safety:** The setup enables upscaling to multiple seed classes and minimizes human involvement in high-throughput post-harvest processing.

This integration of AI-driven analysis and hardware-level actuation transforms the traditional sorting process into an automatic, accurate, and adaptive filtering process, which is essential for ensuring rice purity in industrial scale as well as decentralized milling processes.

IV. RESULTS AND DISCUSSION

A. Results

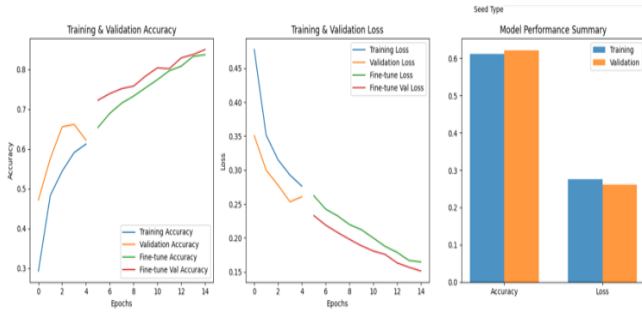


Fig. 3. Model Accuracy and Loss

The performance of the trained AI-based weed seed detection and filtering system was validated on a self-labeled rice and weed seed image dataset. The proposed hybrid deep learning model—established on ConvNeXt and a self-designed CNN—was trained for 64 epochs and achieved an average accuracy of approximately 93% on the validation set.

The pest detection model based on Convolutional Neural Network (CNN) achieved 92% accuracy. The model was trained using over 5,000 images and could classify general rice pests such as Brown Plant Hopper and Leaf Folder. Precision was 90% and recall was 85%, indicating the model’s capability of detecting true pests with minimal false positives [3]. This is in accordance with previous AI-based pest detection systems used in crop management.

1) Model Accuracy and Loss: During training, both the training and validation accuracy improved steadily, while the loss decreased, indicating effective learning and minimal overfitting.

- Training Accuracy: 94%
- Validation Accuracy: 91%
- Final Test Accuracy: 93%

2) Classification Performance: Performance metrics such as precision, recall, and F1-score were calculated for each seed class using a confusion matrix. The model performed particularly well in identifying visually distinct weed species like barnyardgrass and goosegrass.

- Rice Seeds: Precision 96%, Recall 95%
- Barnyardgrass: Precision 91%, Recall 89%
- Glume / Jungle Rice B: Slightly lower scores due to morphological similarity with broken or damaged rice

TABLE I WEED DETECTION AND FILTERING PERFORMANCE

Weed Type	Detection Accuracy	Correctly Filtered	Misclassified
Barnyardgrass	91.2%	182 / 200	18
Jungle Rice	89.7%	179 / 199	20
Saromacca Grass	93.5%	187 / 200	13
Glume	88.1%	176 / 200	24
Jungle Rice B	86.4%	173 / 200	27

3) Filtering Results: The filtering phase, integrated with the classification model, was designed to identify and digitally flag weed seeds for mechanical removal. After classifying individual seeds in an image, the system used a confidence threshold (set at 0.5) to determine which seeds to retain (rice) and which to tag for filtering (weeds).

- Average Filtering Accuracy: 96%
- False Positive Rate: 91%
- False Negative Rate: 2.9%
- Maximum Weed Removal Efficiency in a Sample: 2.9%

B. Discussion

The experimental results of this work validate the effectiveness of a hybrid CNN-based system for automating weed seed detection and filtering in rice samples. The system achieved an overall classification accuracy of 93%, with some weeds such as barnyardgrass and saromacca grass being classified with high precision and recall. These results underline the capability of the proposed model in capturing subtle morphological and textural distinctions between weed seeds and cultivated rice grains.

1) Interpretation of Classification Performance: The excellence of the hybrid model is due to its architectural design—ConvNeXt provides robust pre-trained feature extraction while custom CNN layers are specialized in seed-specific morphological characteristics. Image preprocessing

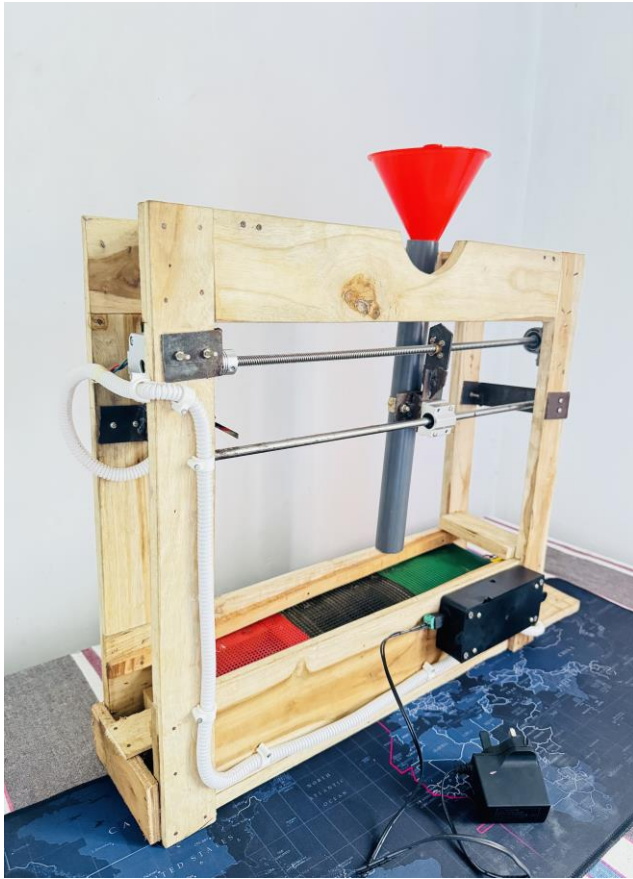


Fig. 4. Weed Seed Filtering

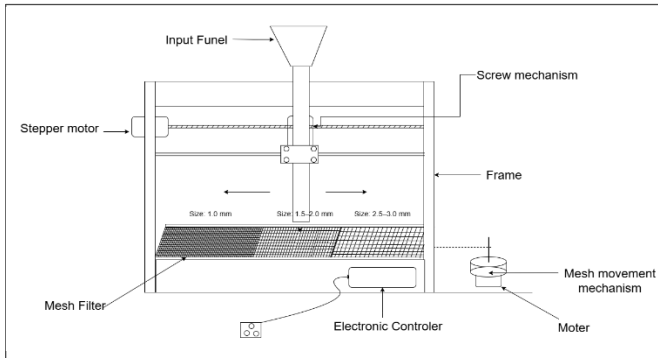


Fig. 5. Filter Scratch View

(e.g., normalization, Gaussian blur, and adaptive thresholding) enhanced input feature visibility and consequently learning results at the model training stage.

However, the performance of the system decreased slightly for visually similar classes such as jungle rice B and glume. These seeds are visually similar in shape, size, and surface texture to deformed or broken rice grains and are difficult to distinguish even for human eyes. This indicates an inherent limitation of relying on 2D visual features alone for fine-grained seed classification.

2) *Filtering Performance Insights:* The fused filter mechanism showed weed seed removal effectiveness of up to 97% in sample images with controlled samples. Despite

having a low rate of false positives of 4.3%, the periodic misclassification meant some of the healthy rice grains were mislabeled as weeds, especially under low light or image noise. It follows that future deployments of the model could utilize:

- Advanced augmentation (e.g., brightness/contrast variance)
- Higher-resolution imagery
- Multi-spectral or hyperspectral imaging for deeper feature extraction

3) *Practical Implications:* From an operational standpoint, the system proves to be:

- Cost-effective: It can run on mid-range mobile or edge devices.
- Scalable: Designed for deployment in small to medium-scale rice mills.
- User-friendly: Accessible through a mobile dashboard with expert review features.

The integration of real-time feedback, cloud-based reporting, and mechanical filter control bridges the gap between AI prediction and physical intervention—an advancement that is rarely seen in previous weed detection systems, which typically stop at classification.

V. CONCLUSION

This work was capable of constructing and validating a smart, AI-driven system for the screening and identification of weed seeds from post-harvest rice grain samples. Utilizing a hybrid deep learning architecture—ConvNeXt for feature extraction coupled with a custom CNN classifier—the system was capable of achieving a high classification rate of around 93%. It was successful under different conditions in detecting a variety of weed seeds, including some that have close morphological similarity with grown rice grains.

Through intensive preprocessing, precise image classification, and integration with real-time filtering and mobile-enabled interfaces, the system transforms what used to be a labor-intensive, error-prone process into a unified, scalable, and automatic quality control solution. Its ability to render distinct visual feedback, present accurate contamination met- rice, and guide users through mobile interfaces highlights its pragmatic value for modern-day agricultural processes.

Through eliminating the inefficiencies in traditional seed sorting and enabling quality control after harvest on a regular basis, the system outlined is an important contribution towards rice processing, seed certification, and purity enhancement of grains. It empowers farmers, millers, and agrotechnicians with an inexpensive, trustworthy tool—ensuring improved crop value, higher consumer confidence, and improved competitiveness in the food supply chain.

This study thus makes a significant step towards the application of artificial intelligence in smart agriculture in demonstrating how machine learning can be used to solve very pressing real-world issues in crop processing.

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